COMP10062: Assignment 4

Mohawk College, Fall 2025

The Assignment

This assignment is about using arrays. You will define several classes – one to represent a playing card with a suit and a rank, and one to represent a deck of cards (these are the **model** classes). Then you will repeatedly shuffle and deal hands, and create a histogram of the results. You will also define a **view** class for this assignment which will use a `main` method and standard input and output.

The Deck of Cards (model)

A `Card` object has a rank and a suit, both of which are positive, non-zero integers. The suit will be displayed as a capital letter starting with A. The rank and suit are specified when a `Card` is created and cannot be changed. constructor A `Card` can report its rank and suit, ~~and its value. The value of a card is its rank multiplied by its suit.~~ It has a `toString()` method that reports its rank and suit. That's all it can do.

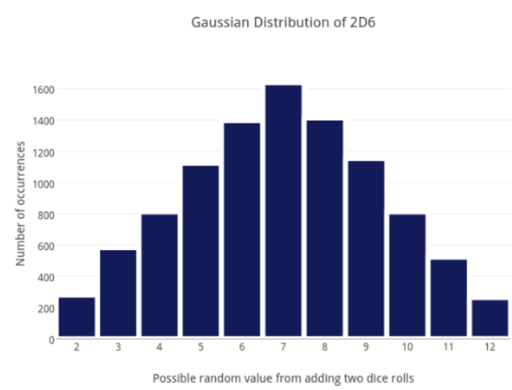
You can convert numbers to capital letters with the following code:

String letter = "" + (char)((int)'A' + number);

A `DeckOfCards` object holds a set of `Card` objects in an array. ~~The size of the deck is determined by two integers passed to the constructor when it is created.~~ The first integer specifies the maximum rank, and the second specifies the number of suits. The deck then contains one card of each rank * suit combination.

For example: If there are 3 suits, and the maximum rank is 6, there will be 18 cards in the deck (3 x 6 = 18). This constructor creates the Card objects and stores them in the array. A DeckOfCards can shuffle itself (which it does by randomly swapping pairs of cards in the array), can report its size, the minimum and maximum card value in the deck, and can "deal" the top `n` cards by placing them in an array and returning them. It has a toString() method that that reports the size of the deck, the minimum and maximum values, and the top card. It also has a histogram method, described below. That's all it can do.

The Histogram Method



The histogram method of `DeckOfCards` accepts a parameter that specifies the number of cards in a hand. Then it shuffles and deals the cards 100,000 times by calling its own shuffle and deal methods. For each deal, it adds up the total value of all the cards in the hand and uses an array of counters to record the number of times each total comes up. For example, if the sum of the values of the cards dealt is 12, you should add 1 to array element 12. Then it returns the histogram array.

See HistogramExample.java in this weeks examples for some code that might help.

Design and Implementation

The exact details of the implementation and interface of these two classes is up to you, but you must respect the specifications given above, and you must create and hand in a UML class diagram to represent the View class, Card class and DeckOfCards and the association relationship between the classes. Use draw.io and submit your drawing as a .jpeg file.

The Main Method (View)

The `main` method should start by asking the user to enter the number of suits and maximum rank. Then it should create the deck of cards and print it to the screen. Then it should present a menu in a loop that allows them to shuffle once, deal one hand, or shuffle and deal 100,000 hands. They choose the size of the hands dealt.

If they choose to deal a single hand, show the result by printing the cards they got. If choose 100,000 hands, call the histogram method described above, and print the non-zero elements of the array, as shown in the example output.

Example Output

Below is an example output of the program. User input is boldfaced and red.

```
How many suits? 2
How many ranks? 3

Deck of 6 cards: low = 1 high = 6 top = Card SAR1
1=shuffle, 2=deal 1 hand, 3=deal 100000 times, 4=quit: 1

Deck of 6 cards: low = 1 high = 6 top = Card SBR2
1=shuffle, 2=deal 1 hand, 3=deal 100000 times, 4=quit: 2
How many cards? 3

Card SAR1 Card SBR1 Card SAR3

Deck of 6 cards: low = 1 high = 6 top = Card SAR1
1=shuffle, 2=deal 1 hand, 3=deal 100000 times, 4=quit: 3
How many cards? 3

     5:  5001
     6:  9981
     7: 14801
     8:  9968
     9: 20099
    10:  9918
    11: 15098
    12: 10078
    13:  5056

Deck of 6 cards: low = 1 high = 6 top = Card SBR2
1=shuffle, 2=deal 1 hand, 3=deal 100000 times, 4=quit: 4
```

BYE!

Optional Extra: Scale the numbers down and print asterisks to represent the quantities.

```
 5:  5001  *****
 6:  9981  **********
 7: 14801  ***************
 8:  9968  **********
 9: 20099  ********************
10:  9918  **********
11: 15098  ***************
12: 10078  **********
13:  5056  *****
```

Validation

Why is 5 the smallest # for this example, and 13 is the largest number? Are these numbers correct? Why isn't 3 the smallest # if you can pick 3 cards? You should be able to explain this example.

Submission

See canvas for submission and evaluation instructions.